CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

Second Semester of M. Sc. (Mathematics) Examination May 2018

MA722 : Differential Geometry

Date : 08/05/2018, Tuesday   Time: 10:00 a.m. to 10:45 a.m.   Total marks 20

Multiple Choice Questions

Instructions:

1. Tick the correct answer only in this sheet.
2. You can use only non-programmable calculator. No other gadget is allowed in the examination hall.
3. Maximum time to answer Multiple Choice Questions will be 45 minutes.
4. Each question carries one mark.
5. All notations and terminologies are standard.

Q-I Choose the correct answer from the given options in the following: [20]

(1) Which of the following curve $\bar{\gamma}: R \to R^3$ is a regular curve ?

(a) $\bar{\gamma}(t) = (t^3, t^2, 0)$ (b) $\bar{\gamma}(t) = (t^2, 0,0)$

(c) $\bar{\gamma}(t) = (t^3, 0,0)$ (d) $\bar{\gamma}(t) = (t, 0,0)$

(2) The reparametrization of the curve $\bar{\gamma}(t) = (r\, cost, r\, sint, 0),\ r > 0$ by arc length s is

(a) $\vec{\beta}(s) = \left(-\,sin\frac{s}{r}, cos\frac{s}{r}, 0\right)$ (b) $\vec{\beta}(s) = \left(-\frac{1}{r}\,cos\frac{s}{r}, -\frac{1}{r}sin\frac{s}{r}, 0\right)$

(c) $\vec{\beta}(s) = \left(r\,cos\frac{s}{r}, r\,sin\frac{s}{r}, 0\right)$ (d) $\vec{\beta}(s) = \left(-r\,cos\frac{s}{r}, -r\,sin\frac{s}{r}, 0\right)$

(3) Which of the following curves represent a circular helix ?
(a) $\bar{\gamma}(t) = (t^3, t^2, t)$
(b) $\bar{\gamma}(t) = (r\, cost, r\, sint, ht), r, h > 0$
(c) $\bar{\gamma}(t) = (2\, sint, cost, 0)$
(d) $\bar{\gamma}(t) = \left(t, \frac{t^2}{2}, 0\right)$

(4) Which of the following smooth maps preserve the first fundamental form?
(a) local isometry (b) local diffeomorphism
(c) diffeomorphism (d) None of these

(5) The torsion τ of a curve $\bar{\gamma}$ is defined as _________.
(a) $\tau(s) = -\langle \overline{B'}(s), \overline{N}(s) \rangle$ (b) $\tau(s) = -\langle \overline{T'}(s), \overline{N}(s) \rangle$
(c) $\tau(s) = -\langle \overline{N'}(s), \overline{N}(s) \rangle$ (d) $\tau(s) = -\langle \overline{B'}(s), \bar{B}(s) \rangle$

(6) If the torsion τ is zero at all points of a curve, then the curve is a ________.
(a) Circular helix (b) plane curve
(c) Circle (d) None of these

(7) Which of the following equation is **not** of Frenet- Serret equations?
(a) $\dot{\bar{t}} = \mathcal{K}\bar{n}$ (b) $\dot{\bar{b}} = -\tau\bar{n}$
(c) $\dot{\bar{n}} = -\mathcal{K}\bar{n} + \tau\bar{b}$ (d) $\dot{\bar{n}} = -\mathcal{K}\bar{t} + \tau\bar{b}$

(8) The equation of the tangent to the curve $y = x^2$ at (0,0) is _______.
(a) $x = 0$ (b) $y = x$
(c) $y = -x$ (d) $y = 0$

(9) The first fundamental form of the surface $\vec{r} = \vec{r}(u\, cosv\, , u\, sinv\, , f(u))$ is
(a) $ds^2 = (1 + f^2)du^2 + u^2dv^2$ (b) $ds^2 = (1 + f'^2)du^2 + u^2dv^2$
(c) $ds^2 = (1 + f'^2)du^2 + u^2du\, dv$ (d) None of these

(10) Which of the following is true?
(a) Every local isometry is a not local diffeomorphism.
(b) Every local isometry is a local diffeomorphism.
(c) Every local isometry is a diffeomorphism.
(d) Composition of two local isometries is not an isometry.

(11) Which of the following is a Christoffel's symbol for a surface patch σ?
(a) $\Gamma_{11}^1 = \frac{FE_v - 2FF_u + GE_u}{2(EG - F^2)}$ (b) $\Gamma_{11}^1 = \frac{EE_u - 2FF_u + GE_v}{2(EG - F^2)}$
(c) $\Gamma_{11}^2 = \frac{GE_u - 2EF_u + FE_v}{2(EG - F^2)}$ (d) $\Gamma_{11}^2 = \frac{2EE_u - EE_v - FE_u}{2(EG - F^2)}$

(12) Which of the following statement is true for the fundamental magnitudes of second order L, M, N for a plane?
(a) $L = 0\, , M \neq 0\, , N \neq 0$ (b) $L = 0\, , M = 0\, , N = 0$
(c) $L \neq 0\, , M = 0\, , N \neq 0$ (d) $L \neq 0\, , M \neq 0\, , N = 0$

(13) The surface area of $\{(x, y, z) \in \mathbb{R}^3 : x^2 + y^2 = 1, z \in (-1,1)\}$ is __________.
(a) $4\pi^2$ (b) π
(c) 4π (d) 2π

(14) Consider equation $z = \frac{1}{2}[\kappa_1 x^2 + \kappa_2 y^2]$. If one of κ_1 and κ_2 is zero and other is non zero, then the equation represents _______.
(a) Elliptical Paraboloid (b) Hyperbolic Paraboloid
(c) Plane (d) Parabolic Cylinder

(15) The image of the Gauss map for a plane is a ________.

(a) circle
(b) sphere
(c) plane
(d) point

(16) The geodesics on a sphere are _________.

(a) All circles
(b) Helices
(c) Great circles only
(d) None of these

(17) Let σ be a surface patch and let $\overline{t_1}$ and $\overline{t_2}$ be the unit principle directions at a point p. Then which of the following is an orthonormal basis of T_pS?

(a) $\{\sigma_u, \overline{t_1}\}$
(b) $\{\overline{t_1}, \overline{t_2}\}$
(c) $\{\sigma_v, \overline{t_1}\}$
(d) None of these

(18) Which of the following is not true for a unit speed curve $\bar{\gamma}$ on surface S?

(a) $\kappa_n = -\kappa \cos\psi$
(b) $\kappa_n = \kappa \cos\psi$
(c) $\kappa_n = \ddot{\bar{\gamma}}\bar{N}$
(d) $\kappa_g = \ddot{\bar{\gamma}}(\bar{N} \times \dot{\bar{\gamma}})$

(19) The sum of interior angles of a regular $n - gon$ on a plane is ______.

(a) $(n+2)\pi$
(b) $> \pi$
(c) $(n-2)\pi$
(d) π

(20) Which of the following is analogus to the *Fundamental Theorem of space curves*?

(a) Gauss-Bonnet theorem
(b) Mainardi-Codazzi equations
(c) Bonnet theorem
(d) Gauss equations

Candidate's Seat No.___________

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY
Second Semester of M. Sc. (Mathematics) Examination May 2018
MA722 : Differential Geometry
Date: 08/05/2018,Tuesday Time 10:45 a.m. to 1:00 p.m. Total marks of Section I & II 50

Instructions:

6. Section I and Section II must be written in separate answer books.
7. You can use only non-programmable calculator. No other gadget is allowed in the examination hall.
8. Figures to the right indicate marks.
9. All notations and terminologies are standard.

SECTION - I

Q-II Answer the following:

(A) Attempt any TWO of the following: [8]

(1) Define Weingarten map. Let κ_1 and κ_2 be two principal curvatures at a point p. If κ_n is a normal curvature at the point in a unit direction $\bar{v}$ and $\kappa_1 \geq \kappa_2$. Show that $\kappa_2 \leq \kappa_n \leq \kappa_1$.

(2) (i) Let σ be a surface patch with the unit normal $\bar{N}$. Prove that
$\bar{N}_u \cdot \sigma_u = -L;$ $\bar{N}_u \cdot \sigma_v = \bar{N}_v \cdot \sigma_u = -M$ and $\bar{N}_v \cdot \sigma_v = -N.$

(ii) Show that the sum of interior angles of a triangle on a sphere is greater than π.

(3) (i) Let σ be a surface patch of S and let $p \in S$. Prove that
$\mathcal{H} = \frac{LG-2MF+NE}{2(EG-F^2)}$ and $\mathcal{K} = \frac{LN-M^2}{(EG-F^2)}$.

(ii) State the Bonnet theorem and the Gauss- Bonnet theorem.

(B) Attempt any SIX of the following: [12]

(1) Define reparametrization of a curve $\bar{\gamma}(t)$.
(2) If $\bar{\gamma}$ is a unit speed curve, then prove that $\dot{\bar{\gamma}}(t) \perp \ddot{\bar{\gamma}}(t)$ for all t.
(3) Define *Torsion* and *arc length* of a curve.
(4) Let $\bar{\gamma}$ be a regular curve in R^3 with nowhere vanishing curvature. If its torsion is identically zero, then show that $\bar{\gamma}$ is planner.
(5) Prove that every local isometry is a conformal map.
(6) Show that the map f^* is a symmetric bilinear map.
(7) Define regular surface and convex curve.
(8) Prove that the principal curvatures are normal curvatures.
(9) If geodesic curvature of a unit speed curve $\bar{\gamma}$ is zero, then show that $\bar{\gamma}$ is geodesic.

SECTION - II

Q-III Answer the following:

(A) Attempt any THREE of the following: [18]

(1) Prove that a local diffeomorphism $f: S_1 \to S_2$ is conformal iff there is a map $\lambda: S_1 \to \mathbb{R}$ such that $f^*\langle v\,, w\rangle_p = \lambda(p)\langle v, w\rangle_p, v, w \in T_pS_1$ for every $p \in S_1$.

(2) Let $\sigma: U \to \mathbb{R}^3$ be a surface patch and let $(u_0, v_0) \in U$. Let $\delta > 0$ be such that $R_\delta = \{(u, v) \in \mathbb{R}^2: (u - u_0)^2 + (v - v_0)^2 < \delta^2\} \subseteq U$. Show that $\lim\limits_{\delta \to 0} \frac{A_{\bar{N}}(R_\delta)}{A_\sigma(R_\delta)} = |\kappa|$, where κ is the Gaussian curvature (at the point $\sigma(u_0, v_0)$) of the surface.

(3) Let $\bar{\gamma}(t) = \sigma\big(u(t), v(t)\big)$ be a unit speed curve on a surface patch $\sigma: U \to \mathbb{R}^3$. Prove that $\bar{\gamma}$ is a geodesic iff $\frac{d}{dt}[E\dot{u} + F\dot{v}] = \frac{1}{2}[E_u\dot{u}^2 + 2F_u\dot{u}\dot{v} + G_u\dot{v}^2]$ and $\frac{d}{dt}[F\dot{u} + G\dot{v}] = \frac{1}{2}[E_v\dot{u}^2 + 2F_v\dot{u}\dot{v} + G_v\dot{v}^2]$.

(4) State and derive Codazzi-Mainardi's equations.

(5) State and derive Gauss's equations.

(B) Attempt any TWO of the following: [12]

(1) (i) State and prove Wirtinger's inequality.

(ii) Show that the Weingarten map W of the surface satisfies the quadratic equation $W^2 - 2\mathcal{H}W + \mathcal{K}I = 0$.

(2) (i) If $\bar{\gamma}$ is a regular curve in $\mathbb{R}^3$, prove that its curvature $\mathcal{K}$ is defined by the formula $\mathcal{K} = \frac{\|\ddot{\bar{\gamma}} \times \dot{\bar{\gamma}}\|}{\|\dot{\bar{\gamma}}\|^3}$.

(ii) Compute signed curvature of $\bar{\gamma}(t) = (t, \cos ht)$.

(3) (i) If $f: S_1 \to S_2$ and $g: S_2 \to S_3$ are smooth maps, prove that $D_p(g \circ f) = D_{f(p)}g \circ D_pf$, for all $p \in S_1$,.

(ii) State and prove Meusnier's theorem without figure.